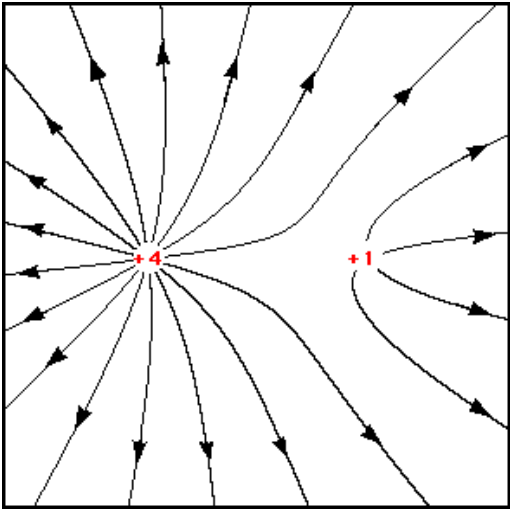


|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>5a</b>    | Zero electric field strengths in sphere A (between $x = 0$ and $x = 1.4$ cm) and in sphere B (between $x = 11.4$ and $x = 12.0$ cm)                                                                                                                                                                                                                                                                                                                                   |
| <b>5bi</b>   | Since the <u>field strength</u> is zero at a point between the spheres OR the <u>electric fields</u> are in <u>opposite directions</u> , the charges on the spheres are of <u>the same</u> sign.                                                                                                                                                                                                                                                                      |
| <b>5bii</b>  | <p>At <math>x = 0.08</math> m, the electric field strength due to sphere A cancels out the electric field strength due to sphere B.</p> $E_A = E_B$ $\frac{Q_A}{4\pi\epsilon_0 (0.08)^2} = \frac{Q_B}{4\pi\epsilon_0 (0.04)^2} \text{ (substitution of distances required for credit of mark)}$ $\frac{Q_A}{Q_B} = \left(\frac{0.08}{0.04}\right)^2 = 4$ <p>Allow estimation from graph, <math>7.8 \text{ cm} &lt; x &lt; 8.2 \text{ cm}</math></p>                   |
| <b>5biii</b> | <p>Diagram</p>  <ul style="list-style-type: none"> <li>• Correct field line direction (either all inwards or all outwards) and shape, and location of neutral point nearer to sphere B.</li> <li>• More field lines radiating out of sphere A with closer spacing than sphere B, perpendicularly from the spheres</li> </ul> <p>E1: Incorrect number and spacing of field lines</p> |